

Weekly Progress Report: Mediation Analysis & Package Validation

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Key updates include: - *Bug Fix*: Identified and resolved an ordering issue in the bootstrap loop where covariate-coefficient mismatching occurred due to duplicate IDs. - *New Feature: Exact Match Mode*: Implemented the `refit = TRUE` parameter in `cmest_pathcomprisk`, allowing for a full non-parametric bootstrap that matches the original script's logic. - *Model Validation*: Performed a head-to-head comparison of the original re-fitting method versus the package's default parametric approach.

Mode Comparison & Validation Results

To ensure the package behaves as expected, I compared two different bootstrap philosophies using a subset of the SHS Mimick data ($N = 500, B = 20$).

1. Parametric/Hybrid (Default)

This mode fits models once and then draws from the parameter distributions (`rmvnorm`) in each iteration. It is significantly faster and computationally stable.

2. Non-parametric (Exact Match)

This mode re-fits all models (mediators, dropout, and outcome) on a new resampled dataset in every iteration. This matches the original script's logic and captures uncertainty in the model-fitting process itself.

Full Dataset Results (N=100,000)

To provide the most accurate assessment, I ran the analysis on the full SHS Mimick dataset (100,000 observations). This allows us to see the stable point estimates and the performance of the Aalen outcome model.

Outcome Model Coefficients (Aalen) The Aalen additive hazards model was fitted on the full dataset to estimate the transition rate to the event of interest:

Covariate	Coefficient (estimate)
Treatment (E)	0.000807
Mediator (M)	0.000151
Covariate (C)	0.001529

Point Estimate Comparison (Full Data) The following mediation effects were calculated using the point estimates from the full dataset (N=100,000).

Effect	Estimate (N=100,000)
Direct Effect (DE)	90.96
Indirect Effect (IEM)	25.79
Competing Risk (IED)	-0.18
Total Effect (TE)	116.57

Note: These point estimates represent the ‘ground truth’ for the current model specification, as they utilize the maximum available information.

Production Run Details

A large-scale production run is currently underway using the following parameters: - Sample Size (N): 7,500 - Bootstrap Samples (B): 1,000 - Mode: Exact Match (`refit = TRUE`) - Estimated Completion: Several hours (running in-process)

Model Definitions (Refit Mode)

The package now handles the re-fitting of these models automatically inside the bootstrap loop:

```
# Example of the new 'refit' call in the package
results <- cmest_pathcomprisk(
  D = D_models,
  mreg = mreg_models,
  mvar = c("M1", "M2"),
  yreg = yreg_model,
  avar = "E",
  data = simData,
  nboot = 1000,
  refit = TRUE,                # Enable Exact Match
  yreg_time = "time_to_event", # Time variable name
  yreg_event = "eventHappened" # Event indicator name
)
```

Next Steps

- Monitor and complete the $N = 7,500$ production run.
- Update the documentation for the `refit` parameter in the `cmcrhazard` package.
- Prepare a final summary report once the large-scale results are available.